LS2103 (BIOPHYSICS) MID-SEMESTER EXAMINATION. 06.09.2025

Time: 90 MINUTES. FULL MARKS: 25 = 24+1 (for neatness)

Answer all questions. Show all necessary steps.

Useful Information:

- Boltzmann constant, $k_B = 1.3806 \times 10^{-23}$ m^2 kg s^{-2} K^{-1}
- Gas constant, $R = 8.3144$ J mol^{-1} K^{-1}
- Planck's constant, $h = 6.6261 \times 10^{-34}$ m^2 kg s^{-1}

Q1. Clearly derive the answers to the following. **(4 x 3 = 12 marks)**

i) Determine the closest *molecular equivalent of (1 RT)* from in (picoNewton-nanometre) at a temperature of 4 °C.

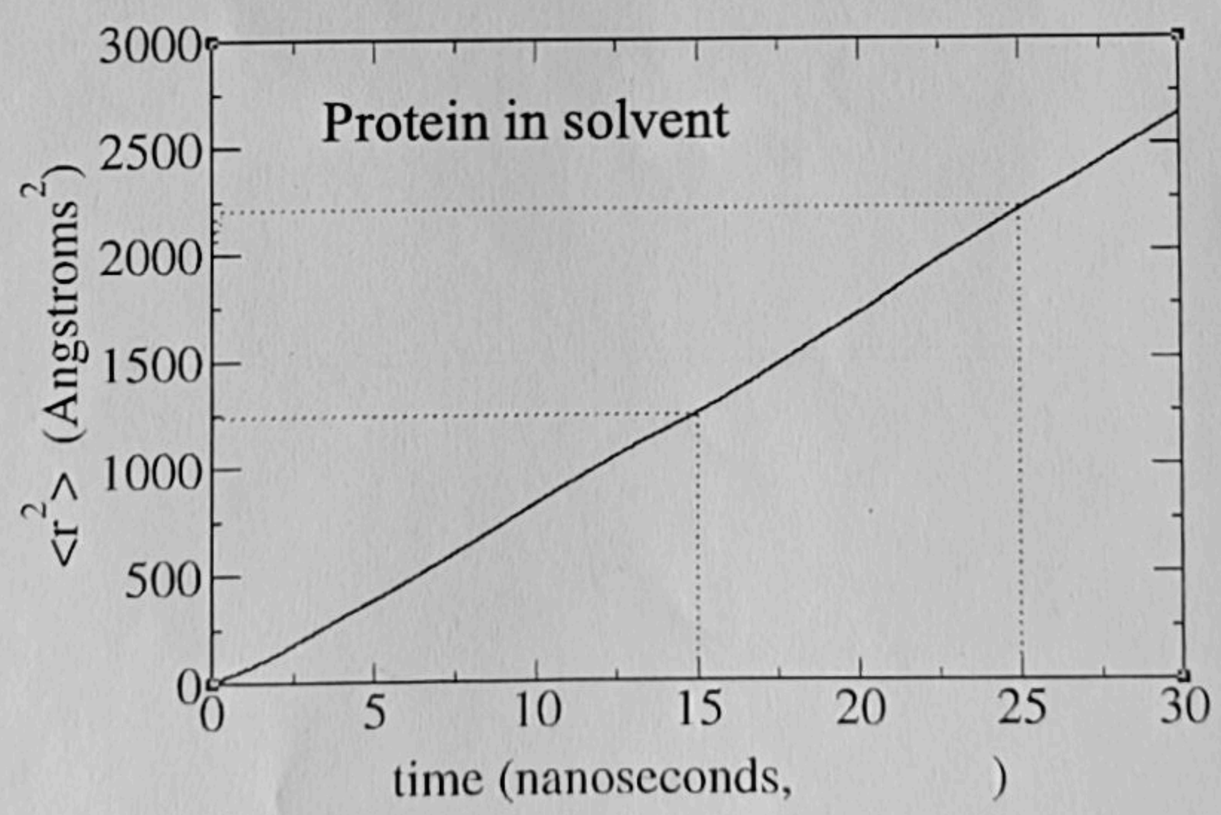
ii) Metabolism rate of a terrestrial vertebrate is measured in two ways: a) number of micromoles of CO_2 exhaled per gram of body weight (say *X*), and b) heat generated per unit body weight (say *Y*).
If [***X* = *B Y***], find the dimensions of '*B*'. Report the SI units of '*B*'.

iii) A study suggests that the (volume) of eukaryotic organisms scales as (surface area)$^{3/2}$, with an average proportionality constant *K*. Derive and schematically depict the plot of log(volume) to log(surface area):

iv) A spherical virus particle of radius '***a*' nanometers** is freely diffusing in an aqueous medium of viscosity 'η' with a diffusion coefficient of **'*D*' *cm*^{2} *second*$^{-1}$**. The temperature is ***T* °C**.
Write down the relationship connecting these quantities. Hence determine the dimensions of 'η', and report its SI units.

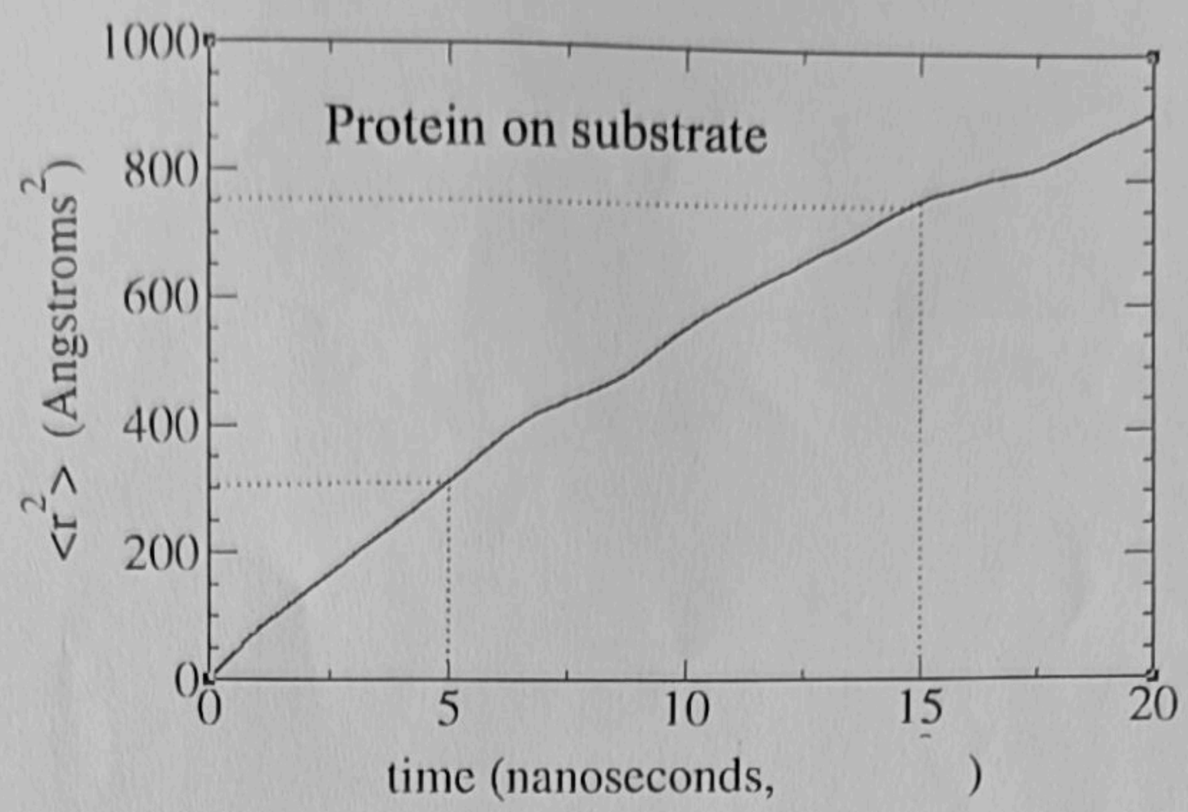
Q2. **(3 + 3 = 6 marks)**

Consider a protein of a certain mass diffusing freely in a solvent, whose mean squared displacement is depicted below:



i) Find the diffusion coefficient of the protein in the solvent in SI units.

ii) An experiment is designed where the same protein, with unchanged diffusivity now moves over a *surface* of unknown characteristics. The mean squared displacement observed in these conditions is shown (see next page).
Obtain the *best estimate* of the spatial dimension (*d*) of the surface. Clearly explain your steps and your reasoning.



Q3. **(2 + 2 + 2 = 6 marks)**

One of the velocity components of a 10 nanoMolar solution of the protein lysozyme is assumed to follow a probability distribution of the following form at room temperature (T Kelvin):

$$P(v_z) = Ae^{-\frac{av_z^2}{b}}$$

i) State the underlying assumptions.

ii) Clearly re-write the variables A, a and b in terms of a specific protein characteristic. All variables must be identified.

iii) Draw a graphical schematic representing the probability distribution.
In the same plot, depict another schematic when the temperature is changed to T_2 = (T + 50 Kelvin).

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